Glossary of Key Terms: Conceptualizing Interaction

Conceptualizing Design

Thinking about what a product should do and how it will work before actually building it.
Example: Planning how a robot waiter should behave before creating it.

Conceptual Model

A high-level idea of how a system works and how users will interact with it.
Example: The "desktop" idea in computers where files and folders behave like real objects.

Problem Space

The set of problems, needs, and challenges that a design is trying to solve.
Example: Difficulty in hiring waiters is a problem space for designing a robot server.

Design Space

The possible solutions and ideas for solving a problem.
Example: Designing a speaking robot vs a simple delivery robot.

Assumption

Something you believe to be true without proof and need to test.
Example: Assuming customers will enjoy talking to a robot waiter.

Claim

A statement that is presented as true but still needs evidence.
Example: Saying a robot waiter will increase restaurant revenue.

Conceptual Analysis

The process of examining ideas, assumptions, and problems carefully.
Example: Questioning whether a robot waiter will be helpful or annoying.

Feasibility

How realistic or possible it is to build something.
Example: Whether current technology can support a talking robot.

Interface

The part of a system that users interact with.
Example: Buttons on a vending machine or a touchscreen.

Interface Metaphor

Using familiar real-world ideas to help users understand a system.
Example: A "shopping cart" on a website works like a real shopping cart.

Metaphor

Describing one thing using another similar idea to make it easier to understand.
Example: "Surfing the web" compares browsing the internet to surfing waves.

Task-Domain Objects

The items that users interact with in a system.
Example: Files and folders in a computer.

Attributes

The properties or characteristics of objects.
Example: File name, size, and type.

Operations

The actions users can perform on objects.

Example: Opening, saving, or deleting a file.

Mapping

The relationship between controls and their effects.
Example: Clicking a folder icon opens it.

Interaction Types

Different ways users interact with a system.
Example: Clicking, speaking, dragging, or exploring.

Instructing

Giving commands to a system to perform tasks.
Example: Clicking “print” to print a document.

Conversing

Interacting with a system as if talking to a human.
Example: Asking a voice assistant for the weather.

Manipulating

Directly interacting with objects by moving or changing them.
Example: Dragging a file into a folder.

Direct Manipulation

A style where users interact directly with visible objects on the screen.
Example: Dragging and resizing images using a mouse.

Exploring

Moving through a physical or virtual environment.
Example: Navigating a virtual city in a game.

Responding

When the system takes action or gives suggestions without being asked.
Example: A fitness app notifying you of your daily step goal.

Feedback

Information a system gives in response to user actions.
Example: A button changing color when clicked.

User Experience (UX)

How a person feels when using a product or system.
Example: A simple app feels easy and enjoyable to use.

Conceptual Framework

A structured way of organizing ideas and understanding a system.
Example: A model showing how users interact with a website.

Coordination

Organizing how different parts of a system or people work together.
Example: A system that manages order placement and delivery.

Hybrid Model

Combining multiple interaction types in one system.
Example: A smartphone that supports touch, voice, and gestures.

Interface Style

The form of interaction used in a system.
Example: Menu-based, voice-based, or graphical interface.

Command Interface

A system where users type commands.

Example: Using a terminal to run programs.

Graphical User Interface (GUI)

An interface with visual elements like icons and windows.
Example: Windows or macOS desktop.

Augmented Reality (AR)

Technology that adds digital objects to the real world.
Example: Seeing a virtual animal in your living room through a phone.

Virtual Reality (VR)

A fully immersive digital environment.
Example: Wearing a headset to explore a virtual world.